

Ekagra Gupta

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EDUCATION

VIT Bhopal University

B.Tech in Computer Science and Engineering (CSDF) — CGPA: 8.5/10

Bal Vidya Mandir School

Class XII (CBSE) — Percentage: 85.6%

Sehore, Madhya Pradesh

2023 – 2027

Lucknow, Uttar Pradesh

2022

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL

Frameworks & Libraries: ReactJS, NodeJS, ExpressJS, Flask, Mediapipe, Scikit-Learn, PyTorch, NumPy, Matplotlib

Databases: MongoDB, MySQL

Security Tools: Wireshark, OWASP ZAP, WebGoat, Selenium, Nmap

Other: Git, PyQt6, Machine Learning, GenAI, FastAPI, REST APIs

PROJECTS

Disability Aid (ISL Gesture Recognition) | *PyTorch, FastAPI, React, MediaPipe*

July 2025 – Mar 2026

- Built a cross-platform desktop app for real-time Indian Sign Language recognition, engineering a 1154-dimensional pipeline using MediaPipe and training a BiLSTM classifier in PyTorch for sub-100ms inference.
- Architected a full-duplex WebSocket pipeline between frontend and FastAPI backend, with a 5-frame majority-vote smoothing engine to filter false positives; packaged as a standalone Windows installer via PyInstaller and Electron.

Red-Team Toolkit | *Python, Selenium, PyQt6, Sockets*

Sep 2024 – Jan 2025

- Built a Python desktop GUI to automate web penetration testing, cutting manual assessment time by ~60%.
- Implemented network analysis features using raw socket programming and whois lookups to map domain infrastructure during reconnaissance phases.

Flight Delay Prediction System | *Python, Scikit-Learn, Flask, Pandas, NumPy*

Jun 2025 – Jul 2025

- Developed an end-to-end ML pipeline covering data preprocessing, feature engineering, and model training across Random Forest, Logistic Regression, and XGBoost classifiers.
- Deployed the best-performing model as a Flask web app enabling real-time delay predictions from user inputs.

EXPERIENCE

The Red Users

Remote

Cybersecurity Intern

Mar 2025 – Apr 2025

- Conducted vulnerability assessments on web applications using OWASP ZAP and WebGoat, identifying and documenting critical risks including SQL Injection, XSS, and CSRF vulnerabilities.
- Documented findings and delivered structured mitigation reports to strengthen client web and network security posture.

CERTIFICATIONS

Microsoft: Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900)

Google: The Bits and Bytes of Computer Networking; Introduction to Generative AI; Generative AI for Developers

IBM: Journey to Cloud: Envisioning Your Solution

NPTEL: Blockchain and its Applications

Stanford: Machine Learning Specialization (Supervised, Unsupervised & Reinforcement Learning) by Andrew Ng

ACHIEVEMENTS & ACTIVITIES

Competitive & Community

- Secured **2nd place** in Mystery Manor, a cybersecurity challenge event organized in AdvITYa, VIT Bhopal.
- Served as **Social Media Team Lead** for Eureka Club and Cyber Warriors Club, VIT Bhopal – managed content strategy and grew engagement across platforms for both clubs.
- Participated in **CodeRed** (NITK) and **KashiCTF** (IIT BHU) – national-level competitive coding and Capture The Flag events.